SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

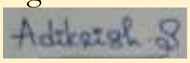
ASSESSMENT TOOL 1 – PSEUDOCODE DESIGN

Course Code	**DSA0404**	**Course Title**	**Fundamentals of Data Science**
CO Assessed	**CO5 - Pseudocode Design**	**Assessment**	**Assessment Tool 1: Pseudocode Design**
Weightage	**30% of CIA**	**Total Marks**	**100**
CO5 Statement	**To understand the concept of supervised and unsupervised methods in machine learning**		
Student Name	**Aditya Krishnan S**	**Reg. No.**	**192424069**
Section/Batch	**2024**	**Date**	**24 August 2026**

Code of Conduct:

I, ADITYA KRISHNAN S/192424069, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Problem	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyse the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Question 1: CART Classification Using Gini Impurity

Problem Statement

Design a CART classification algorithm to select the **best binary split using Gini impurity** and construct a classification tree.

Algorithm

1. Load the dataset.
2. Calculate Gini impurity for possible splits.
3. Select the split with the **lowest Gini impurity**.
4. Divide the data into two branches.
5. Repeat until the stopping condition is reached.
6. Assign the majority class to each leaf.

Pseudocode

```
CART(data):
    if data is pure:
        return Leaf(class)

    best_split = NULL
    lowest_gini = ∞

    for each feature:
        for each possible split:
            divide data into left and right
            calculate weighted Gini impurity

            if Gini < lowest_gini:
                lowest_gini = Gini
                best_split = split

    left, right = split(data, best_split)

    return Node(best_split,
                CART(left),
                CART(right))
```

Output

```
Best Split: Income ≤ ₹40,000
Gini Impurity: 0.00

Classification Tree:
Income ≤ ₹40,000
├── Yes → Reject
└── No  → Approve
```

Question 2: Decision Tree - Customer Purchase Prediction

Problem Statement

Design a Decision Tree to predict whether a customer will purchase a product using **age, income, previous purchases, and browsing history**.

Algorithm

1. Collect customer data.
2. Select the input features.
3. Calculate the best decision split.
4. Build the decision tree recursively.
5. Apply the tree to the new customer.
6. Predict **Purchase / No Purchase**.

Pseudocode

```
Input: Customer data
Features = Age, Income, PreviousPurchases, BrowsingHistory

Build DecisionTree(Features, Purchase)

For new customer:
    start at root
    follow the matching branch
    continue until a leaf
    return predicted class
```

Output

```
Decision Tree:
Income > ₹50,000?
├── Yes → Purchase
└── No
    ├── Browsing = High → Purchase
    └── Browsing = Low  → No Purchase

New Customer:
Age = 28
Income = ₹60,000
Previous Purchases = 2
Browsing = High

Prediction: PURCHASE
```

Question 3: Regression Analysis Workflow

Problem Statement

Design pseudocode for a regression workflow that loads a dataset, identifies independent and dependent variables, trains a regression model, and evaluates predictions.

Algorithm

1. Load the dataset.
2. Select independent variables (X).
3. Select dependent variable (Y).
4. Split data into training and testing sets.
5. Train the regression model.
6. Generate predictions.
7. Evaluate model performance.

Pseudocode

```
Load dataset

Select X = independent variables
Select Y = dependent variable

Split X, Y into training and testing data

Train regression model using X_train, Y_train

Y_pred = model.predict(X_test)

Calculate evaluation metric
Display predictions and performance
```

Output

```
Independent Variables: Age, Income
Dependent Variable: Sales

Model Trained Successfully

Predicted Values: [52.4, 61.8, 74.2]
R² Score: 0.91
```

Question 4: Simple Linear Regression – House Price

Problem Statement

Design pseudocode to predict **house price from house area** using simple linear regression and the least-squares method.

Algorithm

1. Input house area (X) and price (Y).
2. Calculate the regression coefficients.
3. Obtain the equation (Y=a+bX).
4. Substitute the new house area.
5. Predict the house price.

Pseudocode

```
Input X = House Area
Input Y = House Price

Calculate:
    b = Σ(X-X̄)(Y-Ȳ) / Σ(X-X̄)²
    a = Ȳ - bX̄

Regression equation:
    Y = a + bX

Input new area
Calculate predicted price
Display prediction
```

Output

```
Regression Equation:
House Price = 1,50,000 + 4,500 × Area

New House Area = 1,500 sq.ft

Predicted House Price = ₹69,00,000
```

Question 5: Multiple Linear Regression – Student Final Mark

Problem Statement

Design pseudocode to predict a student's **final mark** using attendance, internal marks, assignment marks, and study hours.

Algorithm

1. Load student data.
2. Select the four independent variables.
3. Select final mark as the dependent variable.
4. Train a multiple linear regression model.
5. Input new student's values.
6. Predict the final mark.

Pseudocode

```
Input:
    Attendance
    InternalMarks
    AssignmentMarks
    StudyHours

Y = FinalMark

Train multiple linear regression model

Input new student's values

PredictedMark = model.predict(inputs)

Display PredictedMark
```

Output

```
Model Trained Successfully

New Student:
Attendance = 90%
Internal Marks = 82
Assignment Marks = 88
Study Hours = 4

Predicted Final Mark = 84.6
```